

Physics unit 1 book-in example sums

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1. Calculate the number of electrons in one coulomb of negative charge.
2. Consider two point charges q_1 and q_2 at rest as shown in the figure. They are separated by a distance of 1 m. Calculate the force experienced by the two charges for the following cases:
 - (a) $q_1 = +2 \mu\text{C}$ and $q_2 = +3 \mu\text{C}$
 - (b) $q_1 = +2 \mu\text{C}$ and $q_2 = -3 \mu\text{C}$
 - (c) $q_1 = +2 \mu\text{C}$ and $q_2 = -3 \mu\text{C}$ placed in water ($\epsilon_r = 80$)
3. Two small-sized identical charged insulating spheres, each having mass 1 g are hanging in equilibrium as shown in the figure. The length of each string is 10 cm and the angle is 30° with the vertical. Calculate the magnitude of the charge in each sphere. (Take $g = 10 \text{ m s}^{-2}$)
4. Calculate the electrostatic force and gravitational force between the electron in a hydrogen atom. They are separated by a distance of $5.3 \times 10^{-11} \text{ m}$. The magnitude of charges on the electron and proton is $1.6 \times 10^{-19} \text{ C}$, the mass of electron is $m_e = 9.1 \times 10^{-31} \text{ kg}$ and mass of proton is $m_p = 1.6 \times 10^{-27} \text{ kg}$.
5. Consider four charges q_1, q_2, q_3 and $q_4 = q = +1 \mu\text{C}$ located at four different points on a circle of radius 1 m, shown in the figure. Calculate the total force acting on the charge q_1 due to other three forces.
6. Calculate the electric field at points P, Q for the following two cases, as shown in the figure:
 - (a) A positive point charge $+1 \mu\text{C}$ is placed at the origin
 - (b) A negative point charge $-2 \mu\text{C}$ is placed at the origin
7. Consider the charge configuration as shown in the figure. Calculate the electric field at point A . If an electron is placed at points A , what is the acceleration experienced by this electron? (mass of the electron $= 9.1 \times 10^{-31} \text{ kg}$ and charge of electron $= -1.6 \times 10^{-19} \text{ C}$)
8. A block of mass m carrying a positive charge q is placed on an insulated frictionless inclined plane as shown in the figure. A uniform electric field $\vec{E}$ is applied parallel to the inclined surface such that the block is at rest. Calculate the magnitude of the electric field E .
9. The following pictures depict electric field lines for various charge configurations.
 - (i) In figure (a) identify the signs of two charges and find the ratio $\frac{q_1}{q_2}$
 - (ii) In figure (b), calculate the ratio of two magnitudes and identify the strength of the electric field at three points A, B and C

values of q_1 and q_3

10. Calculate the electric dipole moment for the following charge configurations.
11. A sample of HCl gas is placed in a uniform electric field of magnitude $3 \times 10^4 \text{ N C}^{-1}$. The dipole moment of each HCl molecule is $3.4 \times 10^{-30} \text{ C m}$. Calculate the maximum torque experienced by each HCl molecule.
12. (a) Calculate the electric potentials at points P and Q as shown in the figure
 (b) Suppose the charge $+9 \mu\text{C}$ is replaced by $-9 \mu\text{C}$ then find the electrostatic potentials at points P and Q
 (c) Calculate the work done to bring a test charge $+2 \mu\text{C}$ from infinity to the point Q . Assume the charge $+9 \mu\text{C}$ is held fixed at origin and the $+2 \mu\text{C}$ is brought to the point P .
13. Consider a point charge $+q$ placed at the origin and another point charge $-2q$ placed at a distance of 9 m from the charge $+q$. Determine the point between the two charges at which electric potential is zero.
14. The following figure represents the electric potential as a function of x -coordinate. Plot the corresponding electric field as a function of x .
15. Four charges are arranged at the corners of the square $PQRS$ of side a as shown in the figure.
 (a) In the given configuration, if a charge is brought to the centre of the square, by keeping the four charges fixed at the corners, how much extra work is required for this?
 (b) Suppose a charge q' is brought to the centre of the square, by keeping the four charges fixed at the corners, how much extra work is required for this?
16. A water molecule has an electric dipole moment of $6.3 \times 10^{-30} \text{ C m}$. A sample contains 10^{22} water molecules, with all the dipoles in the external electric field of magnitude $3 \times 10^5 \text{ N C}^{-1}$. How much work is required to rotate all the water molecules from $\theta = 0^\circ$ to 90° ?
17. Calculate the electric flux through rectangle of sides 5 cm and 10 cm kept in the direction of a uniform electric field 100 N C^{-1} . When the angle is 60° , if electric flux through the rectangle is zero, what is the electric field?
18. (i) In figure (a), calculate the electric flux through the closed areas A_1 and A_2 .
 (ii) In figure (b), calculate the electric flux through the cube.
19. A small ball of conducting material having a charge $+q$ and mass m is thrown upward at an angle θ to horizontal with an initial speed v_0 as shown in the figure. There exists a uniform electric field $\vec{E}$ directed along with the gravitational field $\vec{g}$. Calculate the range, maximum height and time of flight of the ball and treat the ball as a point mass.
20. A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm .
 (a) Calculate the capacitance of this capacitor.

(b) If a 10 V battery is connected to the capacitor, what is the charge stored in any one of the plates? (The value of $\epsilon_0 = 8.85 \times 10^{-12} \text{ N}^{-1}\text{m}^{-2}\text{C}^2$)

21. A parallel plate capacitor filled with mica having $\epsilon_r = 5$ is connected to a 10 V battery. The area of parallel plate is 6 cm^2 and separation is 6 mm. Find the capacitance and stored charge. (The dielectric is carefully charged, the battery is disconnected and the dielectric is removed carefully. Calculate the new values of capacitance, stored energy and charge.)
22. Find the equivalent capacitance between P and Q for the configuration shown.
23. Two conducting spheres of radius $r_1 = 8 \text{ cm}$ and $r_2 = 2 \text{ cm}$ are separated by a distance much larger than 8 cm and are connected by a thin conducting wire as shown in the figure. A total charge of $Q = +100 \text{ nC}$ is placed on the bigger sphere. After a fraction of a second, the charge is redistributed and both the spheres attain electrostatic equilibrium.
- (a) Calculate the charge and surface charge density on each sphere.
 - (b) Calculate the electric potential at each sphere.
24. Dielectric strength of air is $3 \times 10^6 \text{ V m}^{-1}$. Suppose the radius of a hollow sphere in the Van de Graaff generator is $R = 0.5 \text{ m}$, calculate the maximum potential difference created by the Van de Graaff generator.